ETH zürich

Mobile Computing

Emerging Technologies
IEEE 802.15 Bluetooth and Zigbee

Stefan Mangold
stefan.mangold@inf.ethz.ch

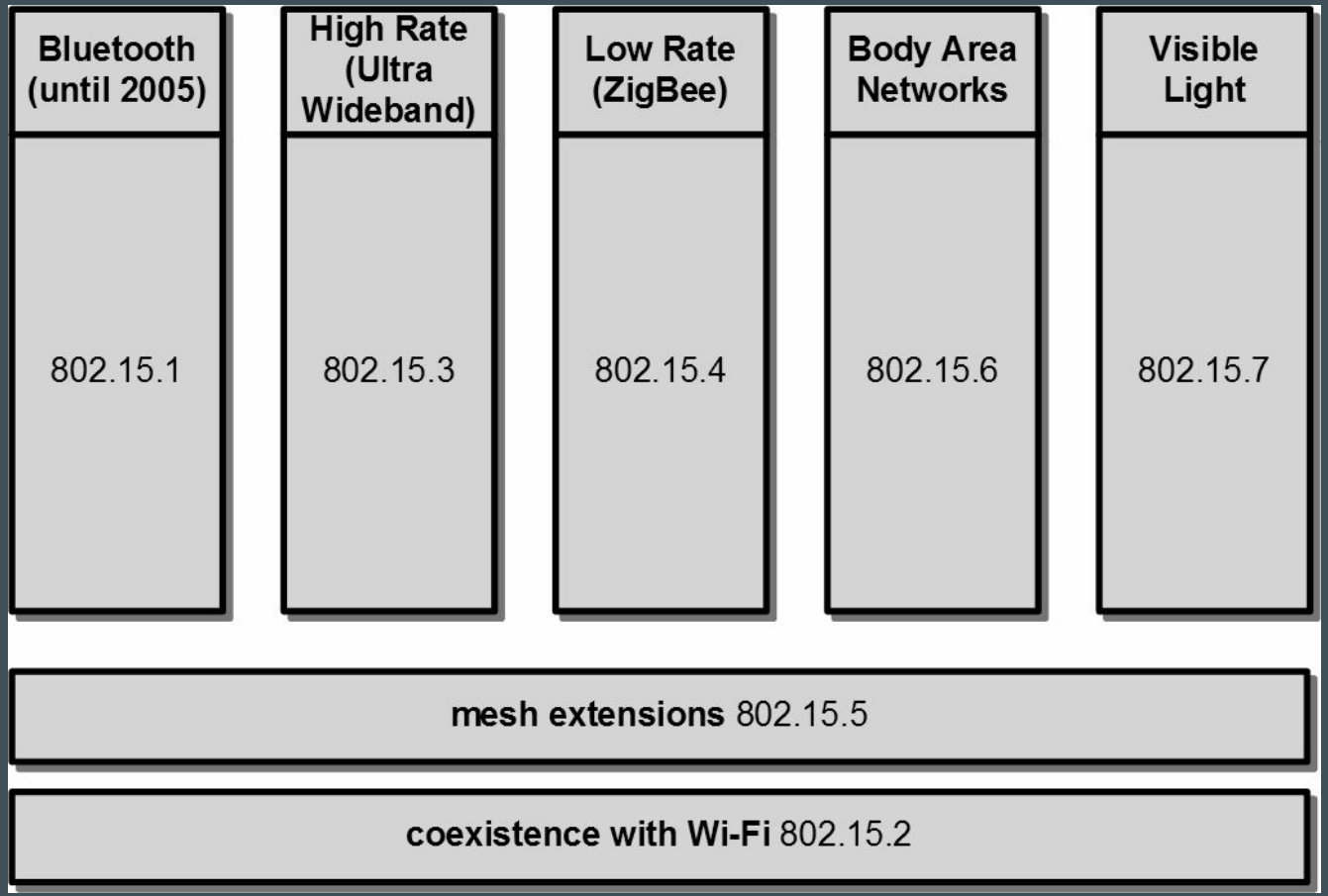
Course Outline

Introduction

1. Data and Signals
2. Data Communication
3. Mobile Systems
4. Mobile Algorithms
5. Mobile AI
6. Cloud Computing

Tournament

IEEE 802.15 Wireless Personal Area Networks



IEEE 802.15 “Personal Area Networks”

- 802.15.1: Bluetooth v1.1
- 802.15.2: Coexistence Mechanisms
- 802.15.3: High Rate WPAN
  - .3a: High Rate Alternative PHY (MB-OFDM vs. DS-UWB)
  - .3b: MAC Amendment
  - .3c: Millimeter Wave Alternative PHY

IEEE 802.15 “Personal Area Networks”

- 802.15.4: Low Rate WPAN → ZigBee
  - .4a: Low Rate Alternative PHY
  - .4b: Revisions and Enhancem.
  - .4c: Alternative PHY for China
  - .4d: Alternative PHY for Japan
  - .4e: MAC for Industrial
  - .4f: Active RFID
  - .4g: Smart Utility Network
- 802.15.5: Mesh
- 802.15.6: Body Area Networks (medical)
- 802.15.7: VLC
- 802.15.8 … 15.10: supporting topics, security

Bluetooth Logo

- In 1997, the specification was named after a Danish king
  - Based on the analogy that it should unite devices the way King Harald Bluetooth united the tribes of Denmark
  - The logo consists of Scandinavian runes for the Viking king's initials, H (ᚼ) and B (ᛒ)

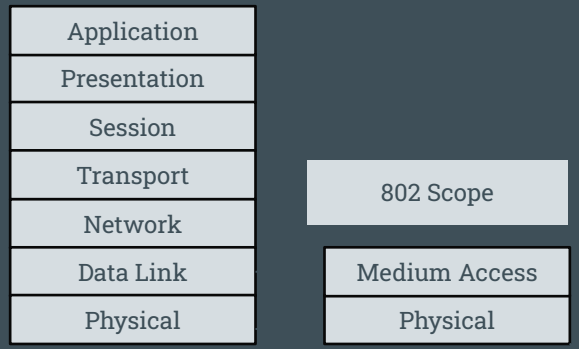


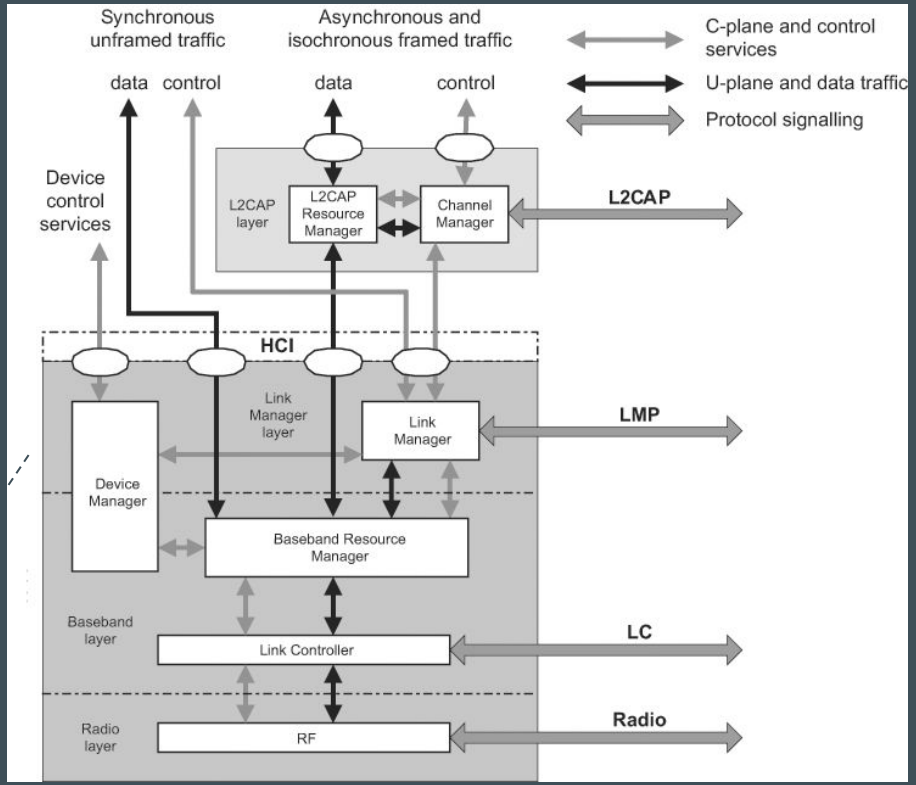
802.15.1 Bluetooth Protocol Stack

- Link Controller (LC)
  - Connection establishment, addressing
  - Packet formatting
  - Timing and power control
- Link Manager Protocol (LMP)
  - Link setup and link management
  - Encryption, authentication
- Logical Link Control & Adaptation Protocol (L2CAP)
  - Adapts upper layer protocols to baseband layer
  - Provides connectionless and connection oriented services

802.15.1 Bluetooth Protocol Stack Illustrated

ISO/OSI Reference Model





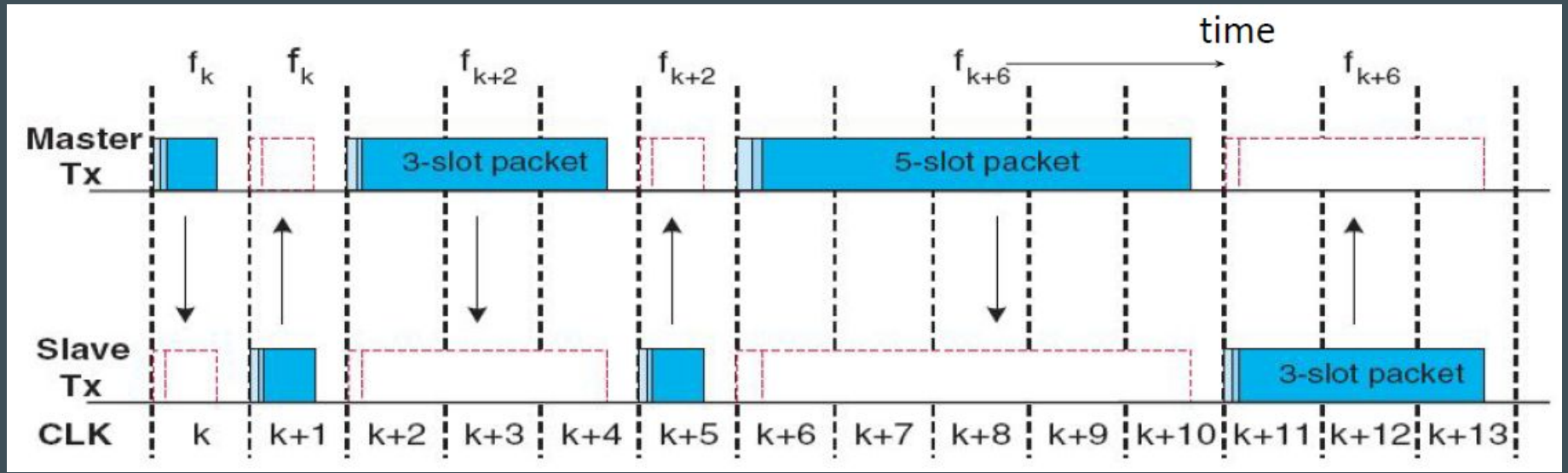
802.15.1 Bluetooth

- Short range, mainly in ISM bands at 2.4 GHz
  - Same as Wi-Fi IEEE 802.11b/g
  - Up to 8 devices in a piconet (1 master and 7 slaves), up to 7 connections
  - Piconets can combine and form so-called scatternets
  - Co-located piconets are linked by sharing one master or one slave device
- Frequency Hopping Spread Spectrum (FHSS)
  - Bluetooth devices hop from one channel to another, pseudo-random, up to 1600 hops per second
  - 2.402 GHz … 2.48 GHz divided into 79 channels with 1 MHz each

802.15.1 Medium Access Control, MAC

- Time division duplex, TDD, time slot length 625 us
  - synchronized clock signals are needed
- Packets are sent on single-hop carrier
  - Frequency remains constant for packet transmission duration
- Frequency Hopping Synchronization (FHS)
  - Fixed and Multi-slot packets (3-slot and 5-slot)
- Master polls the slaves following its own polling scheme
  - Master schedules UL and DL traffic (UL=uplink, DL=downlink)
  - Slave transmits only after it has been polled
  - This polling scheme is not standardized by Bluetooth SIG or 802.15

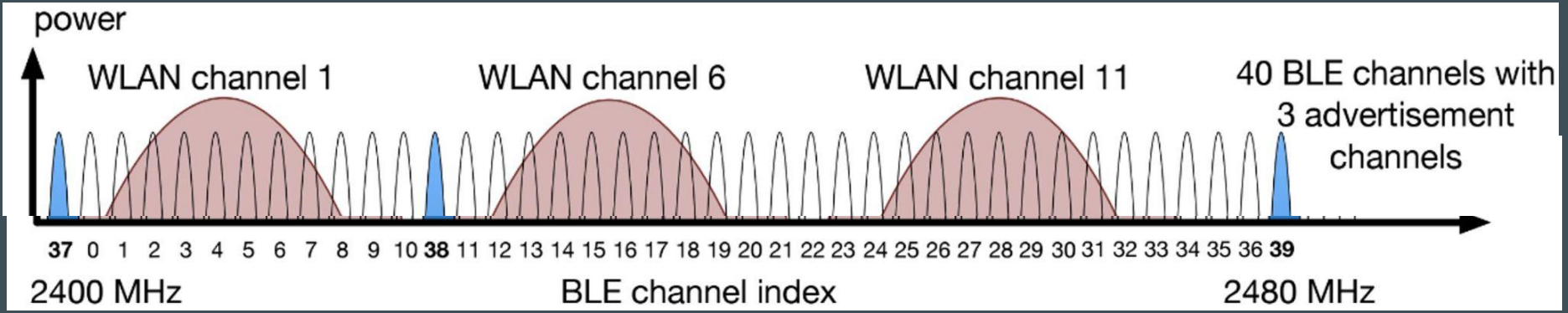
802.15.1: Medium Access Control, MAC



Bluetooth Versions

- Bluetooth 5 is backward compatible with previous Bluetooth versions v1.2 … v4.2
- Bluetooth 5 mainly addresses the need for low power, longer range and higher speed
  - Bluetooth 4.2 reduced power, Bluetooth Low Energy (BLE)
  - Connectionless broadcasting (new: 255 Byte packets) is advantageous for applications like local advertising, guides in shopping mall, for parking
- 1…2 MHz bandwidth, GFSK and QPSK modulation
- Centralized TDMA MAC protocol

BLE Radio Channels



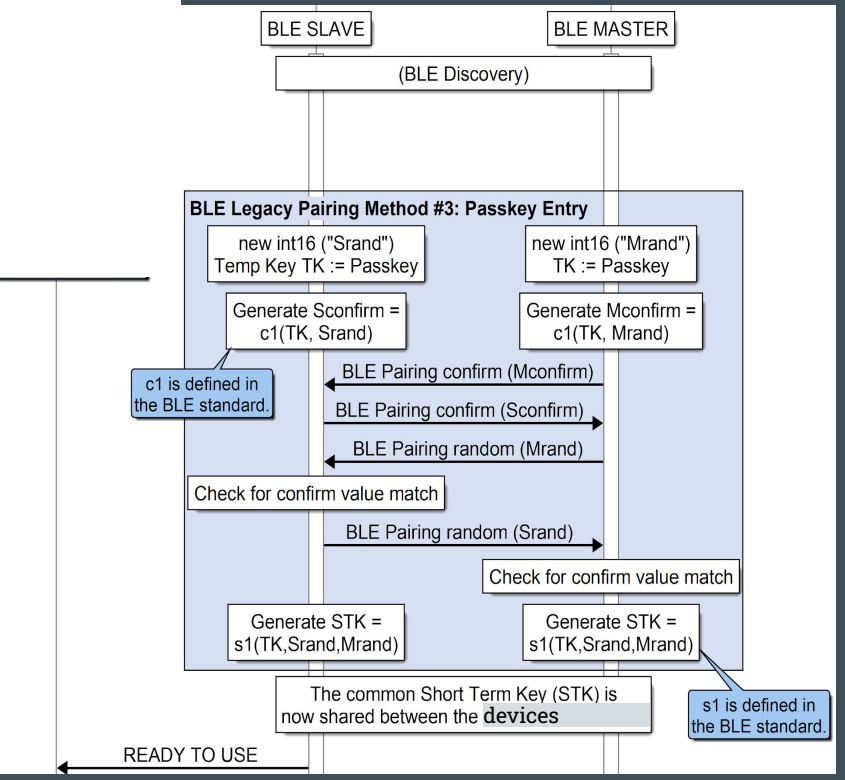
Bluetooth Low Energy (BLE)

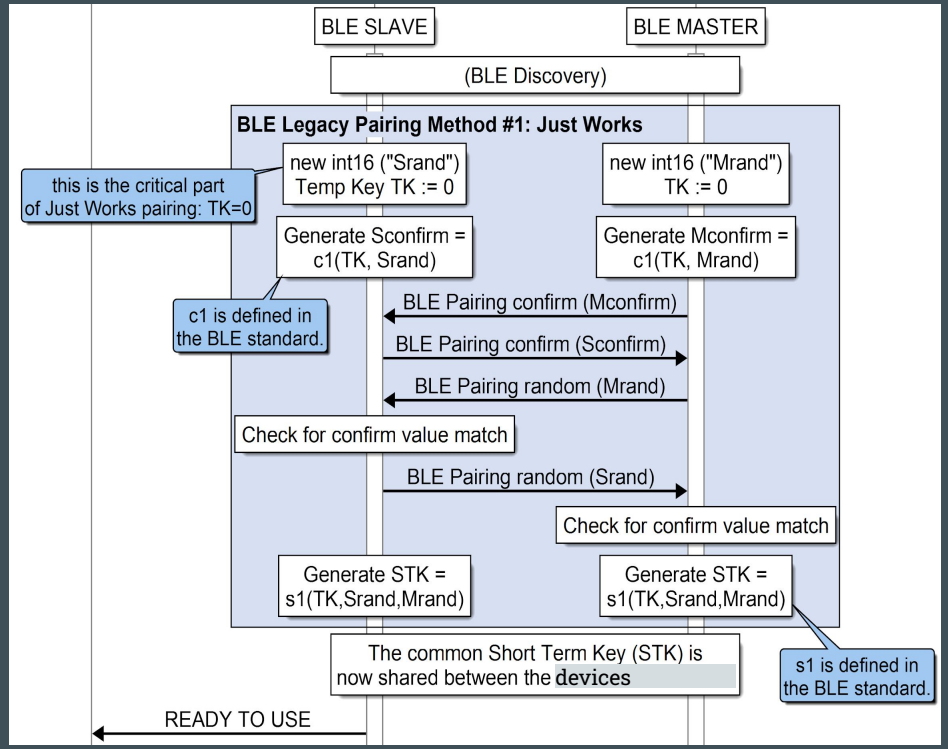
	BLE	Classic BT
Frequency Band	2.4GHz ISM Band	2.4GHz ISM Band
Frequency Channels	3 advertising and 37 data channels, 2 MHz spacing	79 channels, 1 MHz spacing
Freq. Channel Usage	Static on the three advertisement channels, optional adaptive FHSS for data exchange.	Static during advertisement, adaptive FHSS during data exchange
Modulation	GFSK	GFSK, π/4 DQPSK, 8DPSK
Energy Consumption	~0.01x to 0.5x of reference	1 (reference value)
Max Data Rate	125 Kb/s … 2 Mb/s	1 Mb/s … 3 Mb/s
Max Tx Power	1mW (0 dBm) … 100 mW (+20 dBm)	1mW (0 dBm) … 100 mW (+20 dBm)
Network Topologies	Point-to-Point (piconet), Broadcast, Mesh	Point-to-Point (piconet)

BLE Pairing

- Pairing lets the device and phone share a common key
- Not required if secret has already been shared
- Complexity depends on UX and security requirements
- Often used:
  - Just Works
  - Passkey
  - Near Field Communication, NFC (secure but difficult to handle)

BLE Pairing (“Passkey” and “Just Works”)

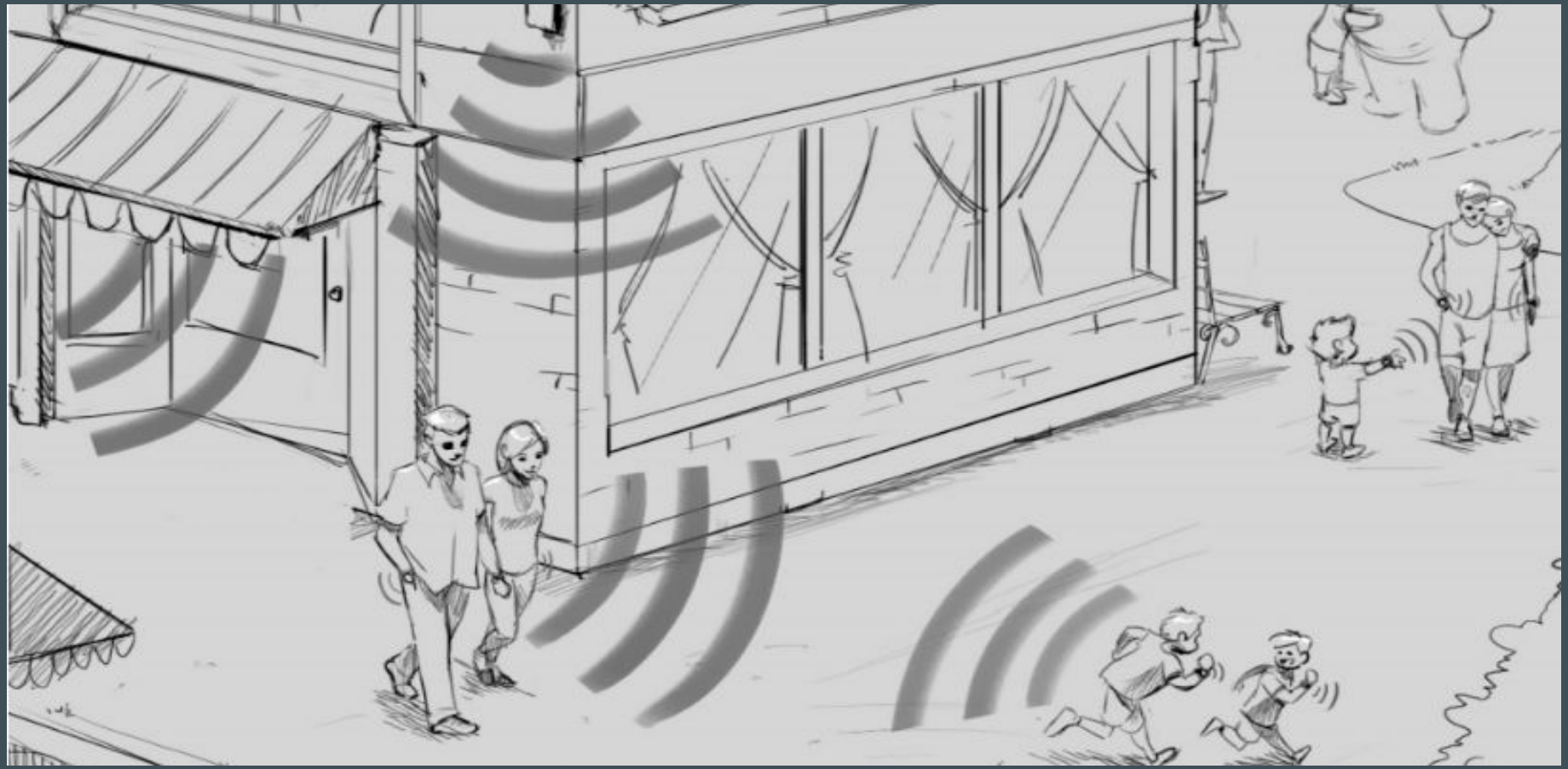




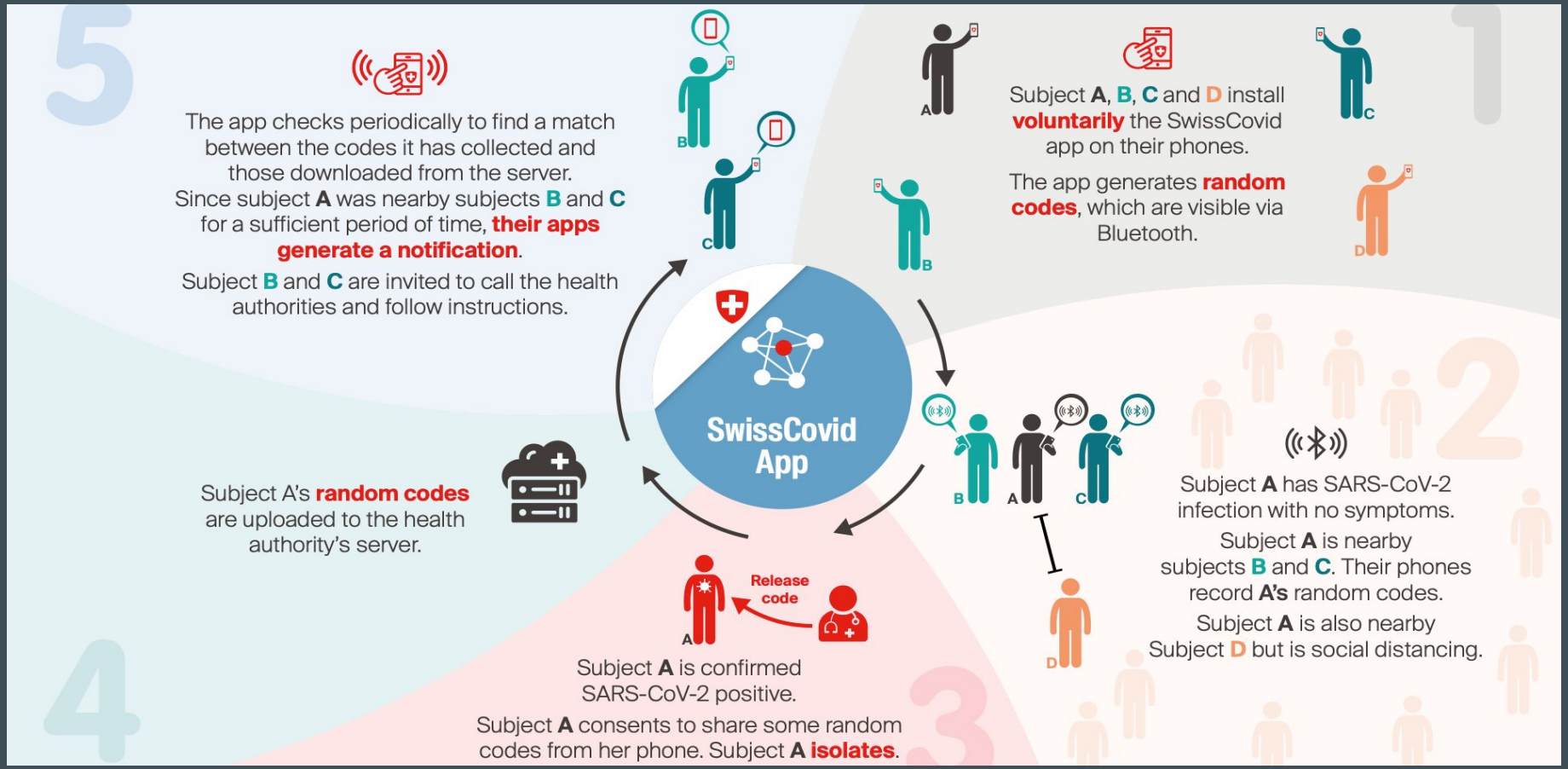
BLE Application: Location Beacons



BLE Application: Device-to-Device (Mesh) Networks



BLE Application: Covid-19 Exposure Notification



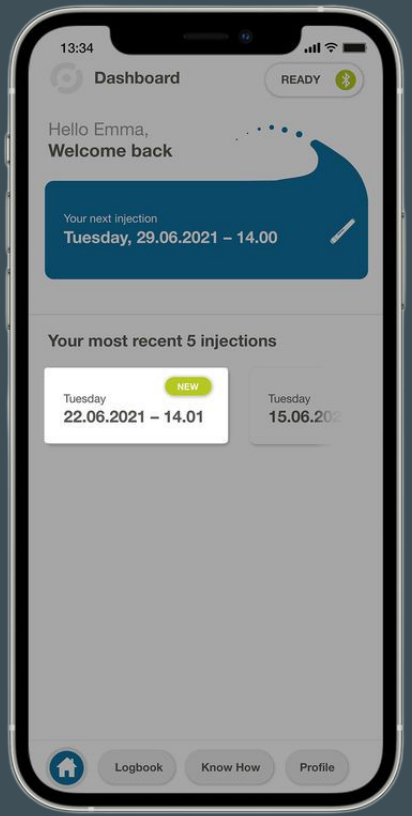
Covid-19 Exposure Notifications

- BLE advertisements are non-connectable and undirected
- BLE advertiser address and associated data are periodically changed (~15min) so that they cannot be linked to a person
- The broadcasting interval ~250 ms
- The notification does not use the location for proximity detection

Use of BLE and VLC in a MedTech Application

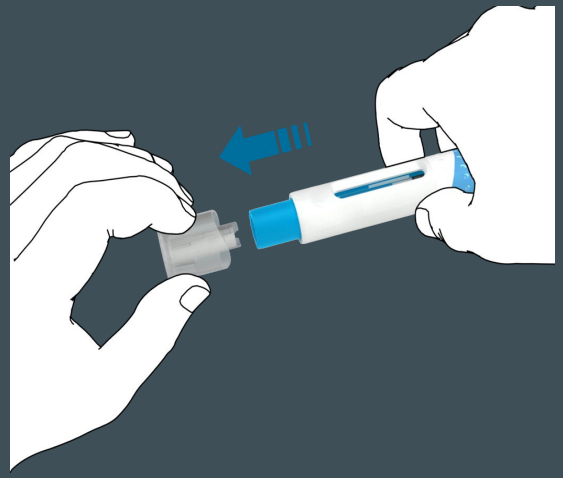
- BLE for Data
- VLC for Activation

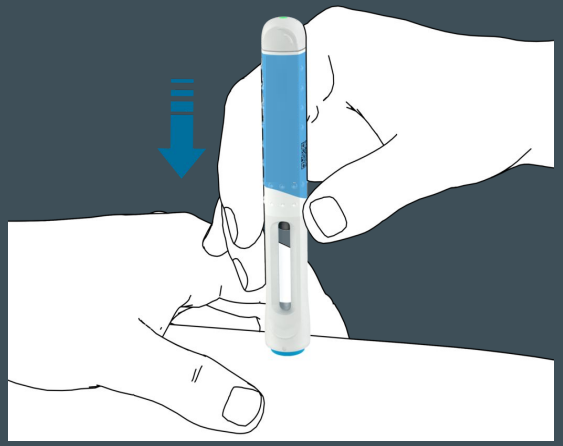


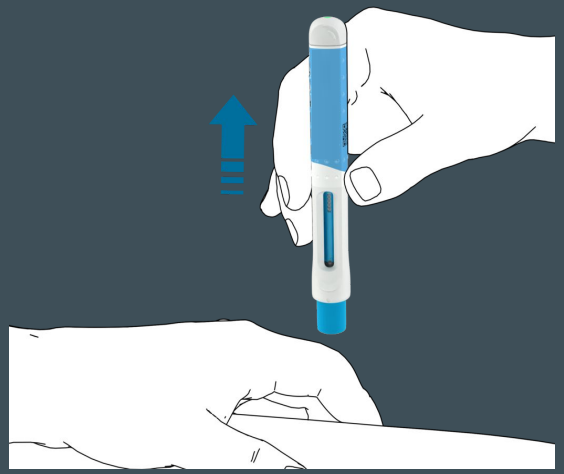


Self-Medication with Low-Complex Autoinjector

- Prefilled injection device
  - Triggered by push-on-skin activation
  - Acoustic confirmation clicks (Source: Ypsomed)







Therapy Management

- Adherence management of self-medication
- Automated injection logging
- Assist patients, caregivers and healthcare professionals



Proximity instead of Pairing

- Transmit the injection-related data to the nearest phones
- A mobile app estimates the distance Based on radio conditions (Bluetooth RSSI)
- App forwards the report to a cloud server
- Cloud resolves report duplications
- No need for additional user steps

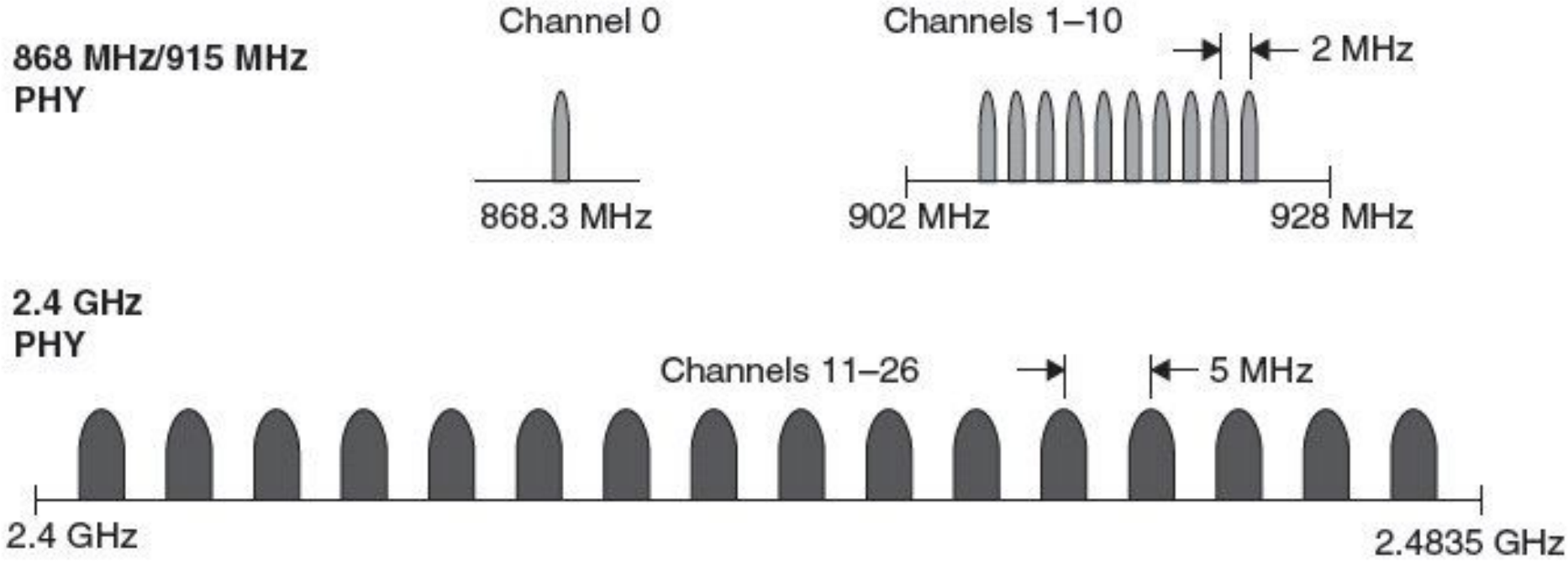


802.15.4 ZigBee

- IEEE Std 802.15.4-2004
- Long battery life (months on AA cell)
  - Networks “form by themselves, scale to large sizes and operate for years without manual intervention”
- Low cost: low device / setup costs
- Up to 65536 nodes
- Data rate 250 kb/s
- 27 frequency channels in two bands
  - 16 channels @ 2.4 GHz, 10 channels @ 915/868 MHz
- Limited data rate and QoS
  - Optimized for power management, mesh support



ZigBee Radio Channels



802.15.4 ZigBee Network Topology

- PAN coordinator (ZigBee Coordinator)
  - Forms root of network tree and might bridge to other networks
  - Introduces basic parameters and stores information about network
  - Only one per network allowed
- Full function device (ZigBee Router)
  - Intermediate router for multi-hop data transmission from other devices
- Reduced function device (ZigBee End Device)
  - Functionality limited to connect to parent node
  - Allows node to be asleep a significant amount of time

802.15.4 ZigBee PHY

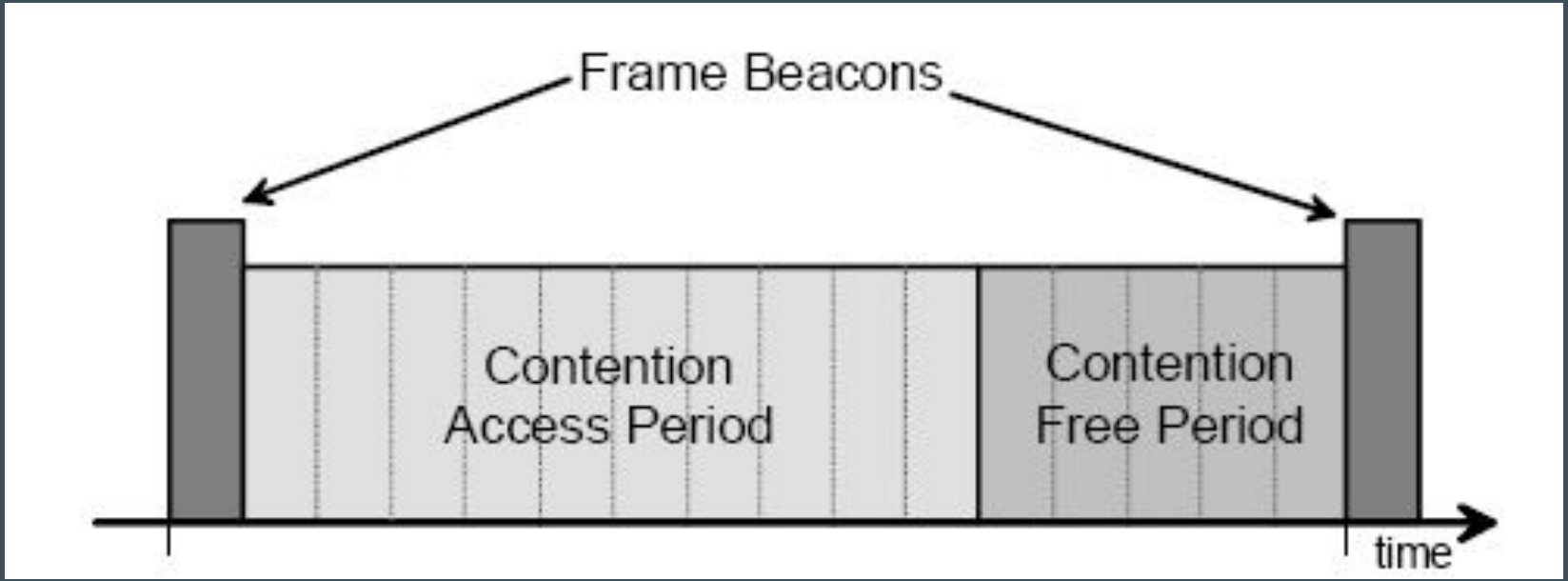
- Direct Sequence Spread Spectrum (DSSS), up to 1 Watt TX

PHY [MHz]	Freq. [MHz]	Chip rate [kchip/s]	Modulation	Bit rate [kb/s]	Symbol rate [ksymbol/s]
868/915	868–868.6	300	BPSK	20	20
868/915	902–928	600	BPSK	40	40
868/915	868–868.6	400	ASK	250	12.5
868/915	902–928	1600	ASK	250	50
868/915	868–868.6	400	O-QPSK	100	25
868/915	902–928	1000	O-QPSK	250	62.5
2450	2400 – 2483.5	2000	O-QPSK	250	62.5

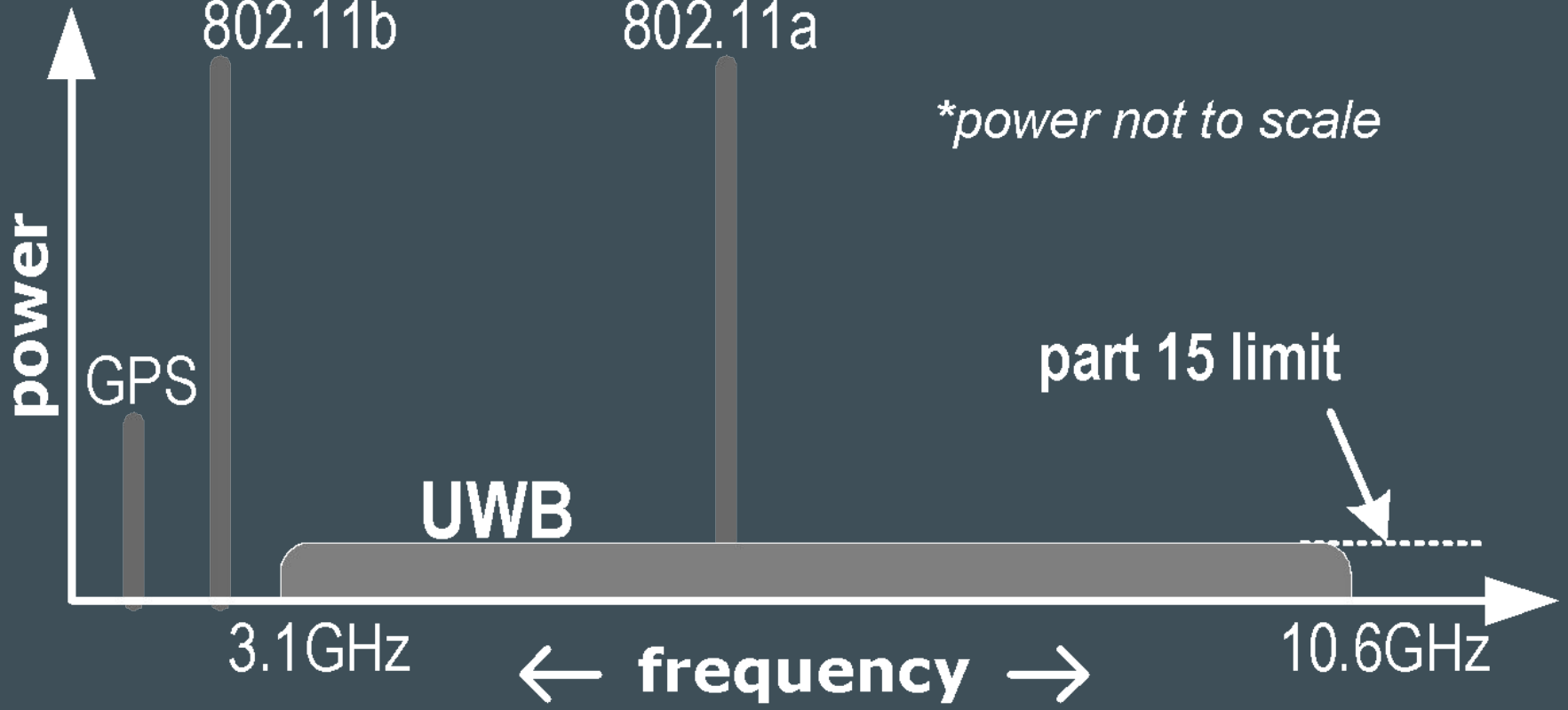
802.15.4 ZigBee MAC

- Variant #1: Non-beacon enabled network
  - Contention Access Period (CAP)
  - Unslotted CSMA / CA
  - Positive acknowledgement for successfully received packets
- Variant #2: Beacon-enabled network
  - CAP + Contention Free Period
  - Network coordinator transmits beacons and sets time slots
  - Slotted CSMA-CA
  - Nodes transmit periodic beacons to confirm their presence
  - Guaranteed Time Slots (GTS) for low latency real-time requirements
  - Nodes may sleep between beacons → extended battery life

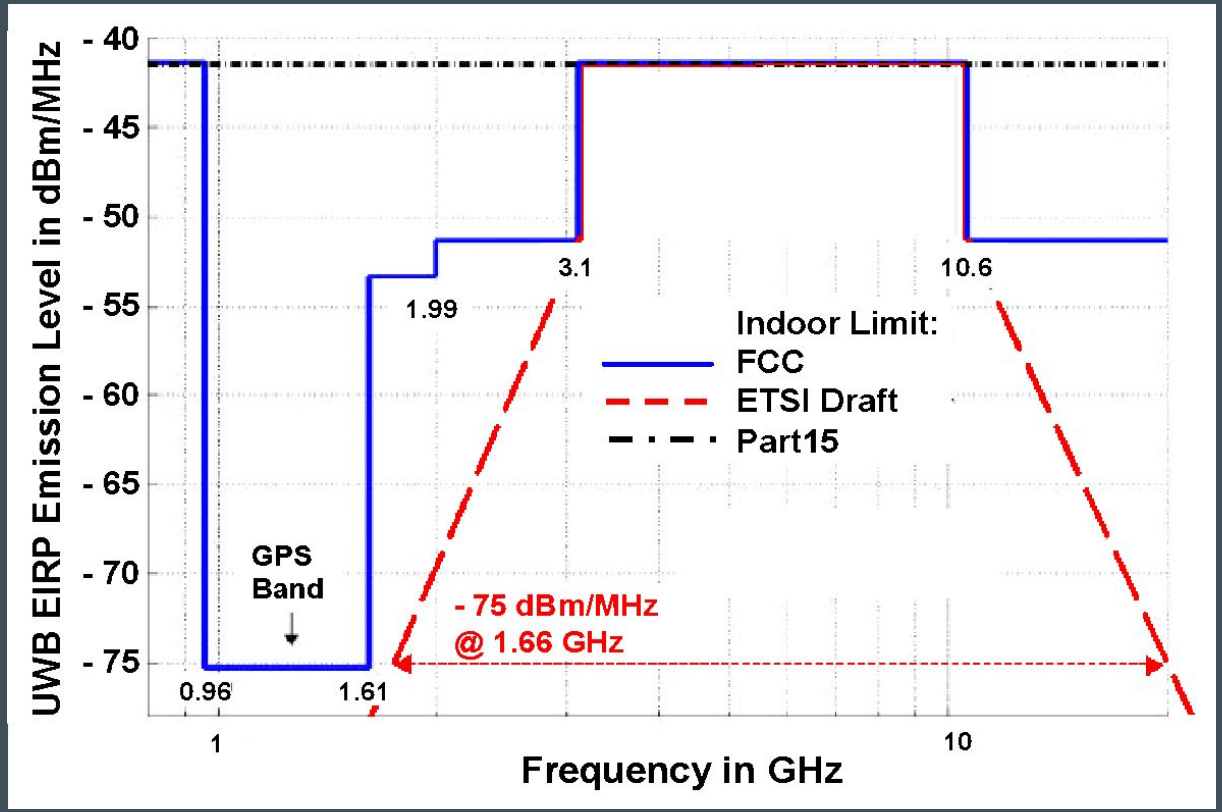
802.15.4 ZigBee MAC illustrated



Ultra Wideband Communications, UWB



UWB Regulation

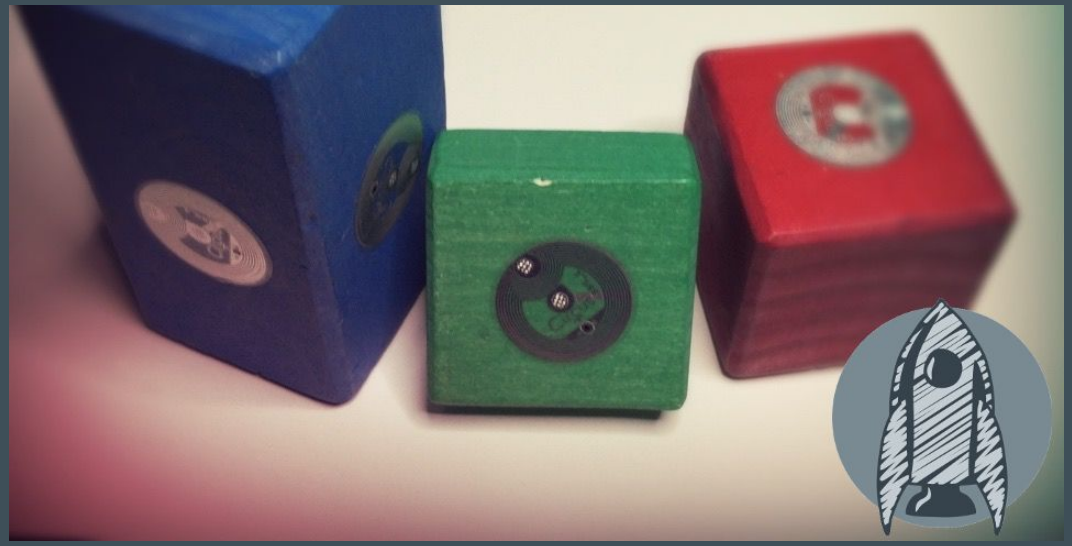


UWB Standard 802.15.3a

	DS-UWB	MB-OFDM
PHY technique	Single-carrier direct sequence spreading	Multiband-OFDM
Frequency management	Two bands (lower band: 3.1 ~ 4.85 GHz Higher band: 6.2 ~ 9.7 GHz)	Four frequency band groups. Each band covers 528 MHz.
Data rates	28, 55, 110, 220, 500, 660, 1000 and 1320 Mb/s	53.3, 80, 106.7, 160, 200, 320, 400 and 480 Mb/s
Typical modulation	BPSK	QPSK
Channel coding	Convolutional coding with interleaving	Convolutional coding with interleaving
Logical channels	Total 12 channels	Total 18 channels through Multiplexing

Near Field Communication, NFC

- Secure communication within a few centimeters
  - No batteries required (similar to RFID, but short range)
  - Often used in logistics, for toys, and consumer products



NFC Technology

- Unlicensed radio frequency ISM band of 13.56 MHz
  - ±7 kHz bandwidth
- Data rates: 106 ... 424 kb/s at up to 10 cm distance
- There are passive and active NFC devices
  - Passive NFC devices (“tags”) store a small amount of data (<10kB)
  - Active NFC devices read from (and write to) other devices or tags, and can also operate as passive NFC tags for peer-to-peer communication
- Standard: ISO/IEC 18000-3

Internet of Things

- Wi-Fi HaLow, BLE, ZigBee
  - Wi-Fi Alliance, Bluetooth Forum
  - Connectivity Standards Alliance, https://csa-iot.org
  - IEEE 802.15.4g - Utility (Smart Meter) Industry
- Proprietary Solution “LoRaWAN”
  - Main inventor: Semtech
- LTE-M, Narrow Band IoT (NB-IoT), Cellular 5G
  - 3GPP, cellular infrastructure

